Winter in Data Science (WiDS)

Exploratory Data Analysis Project

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1. Introduction

This report summarizes my learning journey and project completion during the **Winter in Data Science (WiDS) Mentorship Program 5.0**. Over the course of this winter mentorship, I developed practical data science skills and applied them to a comprehensive Exploratory Data Analysis (EDA) project using Python and Jupyter Notebooks.

Program Objectives Achieved:

- Acquisition of foundational data science and Python programming skills during winter break

- Application of statistical analysis and data visualization techniques to real datasets

- Completion of an end-to-end data analysis project with professional documentation

- Preparation for advanced data science coursework or internships

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2. Learning Journey in WiDS 5.0 Winter Program

2.1 Technical Skill Development

- **Python Programming**: Learned pandas for data manipulation, numpy for numerical computing, and matplotlib/seaborn for data visualization

- **Exploratory Data Analysis**: Applied EDA techniques including data cleaning, transformation, and visualization

- **Version Control**: Used GitHub to manage project files and document progress

- **Jupyter Notebooks**: Created reproducible analysis workflows with clear documentation

2.2 Winter Program Structure & Highlights

[Describe your experience in the winter program. For example:]

- Intensive winter session focused on hands-on learning

- Project-based approach to skill development

- Regular mentorship and feedback sessions

- Peer collaboration opportunities

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3. Project Overview: Restaurant Tips Analysis

3.1 Project Objectives

- Analyze tipping behavior patterns in restaurant transactions

- Explore relationships between bill amount, tip percentage, time, and customer factors

- Practice complete EDA workflow from data loading to insight generation

3.2 Dataset Description

The analysis utilized `tips_analysis_cleaned.csv`, containing restaurant transaction data with features:

- **total_bill**: Final restaurant bill amount

- **tip**: Tip amount given

- **sex**: Gender of bill payer

- **smoker**: Whether party was in smoking section

- **day**: Day of week (Thu-Sun)

- **time**: Lunch or dinner service

- **size**: Size of dining party

3.3 Methodology

The winter project followed a structured EDA workflow:

1. **Data Loading & Inspection**: Initial exploration using pandas

2. **Data Cleaning**: Handling missing values and ensuring data quality

3. **Univariate Analysis**: Understanding individual variable distributions

4. **Bivariate Analysis**: Exploring relationships between variables

5. **Multivariate Analysis**: Examining complex interactions

6. **Visualization**: Creating informative plots using seaborn/matplotlib

7. **Insight Generation**: Drawing conclusions from statistical findings

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4. Analysis & Findings

4.1 Key Statistical Insights

[Copy specific findings from your `analysis_summary.txt` file. Example format:]

- Average tip percentage was approximately 15.8%

- Saturday showed the highest average total bills ($20.44)

- Strong positive correlation (0.68) between total bill and tip amount

- Dinner services generated 30% higher tips than lunch on average

4.2 Visual Discoveries

[Describe 3-4 key visualizations from your Jupyter notebook:]

1. **Histogram of total bills**: Showed right-skewed distribution with most bills between $10-$30

2. **Scatter plot: Bill vs Tip**: Revealed linear relationship with notable outliers

3. **Box plots by day/time**: Illustrated variations in tipping patterns across different times

4. **Heatmap of correlations**: Highlighted relationships between numerical variables

4.3 Notable Patterns from Winter Analysis

- **Pattern 1**: Larger dining parties tend to leave lower percentage tips

- **Pattern 2**: Weekend dinners generate both highest bills and highest total tips

- **Pattern 3**: Consistent tipping percentage (~15-20%) across most customer segments

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5. Challenges & Solutions During Winter Program

5.1 Technical Challenges

- **Challenge 1**: Managing Jupyter notebook dependencies and package versions

  - *Solution*: Created environment.yml file for reproducible environment setup

- **Challenge 2**: Effective visualization for categorical vs numerical data

  - *Solution*: Researched and applied appropriate plot types (violin plots, swarm plots)

5.2 Winter Program Learning Curve

- **Challenge**: Intensive winter timeline requiring rapid skill acquisition

- **Solution**: Focused practice sessions and leveraging mentor office hours

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6. Conclusion & Future Work

6.1 Winter Project Conclusions

This winter project successfully demonstrated the application of EDA techniques to analyze restaurant tipping behavior. The project provided practical experience with the complete data analysis pipeline during the winter break period.

6.2 Skills Gained Through Winter Program

- Complete EDA workflow execution

- Professional documentation with README and final report

- Statistical interpretation and business insight generation

- Technical communication of data findings

6.3 Future Enhancements & Spring Continuation

- Implement predictive modeling for tip amounts using regression

- Expand analysis with additional features (weather, location data)

- Create interactive dashboard using Streamlit or Dash

- Apply learned skills to spring semester projects or internships

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7. References

1. Winter in Data Science (WiDS) 5.0 Program Curriculum

2. Python Data Science Handbook (VanderPlas)

3. GitHub Repository: [Your actual GitHub repo link]

4. Primary Analysis: `EDA final project.ipynb` in attached repository

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8. Appendices

Appendix A: Winter Project Repository Structure

3.3 Detailed Methodology

Phase 1: Data Loading & Initial Inspection

- Used `pd.read_csv()` to load the dataset

- Checked data shape: 244 rows × 7 columns

- Used `.info()` and `.describe()` for initial insights

- Identified data types and missing values

Phase 2: Data Cleaning Process

- Verified no missing values in cleaned dataset

- Converted categorical variables to appropriate dtypes

- Created derived features (tip percentage, etc.)

- Documented all transformations in the notebook

Phase 3: Univariate Analysis Details

- For numerical variables: histograms, box plots, statistics

- For categorical variables: frequency tables, bar charts

- Key observations from each variable's distribution

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4. Add a "Personal Reflection" Section (≈ 0.5 page)

Create **Section 9: Personal Growth & Reflection**:

```markdown

## 9. Personal Growth & Reflection

### 9.1 Before WiDS vs After WiDS

- **Before**: Limited experience with Python data analysis

- **After**: Confident in performing complete EDA and creating reproducible workflows

### 9.2 Most Valuable Learning

- The importance of **data storytelling** through visualization

- How **systematic EDA** reveals hidden patterns

- Value of **documentation** for reproducibility

### 9.3 Application to Academic/Career Goals

- This project strengthens my foundation for [your major] courses
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- Skills directly applicable to internships in data analysis roles
- Built a portfolio piece for graduate school/job applications